

```
from sklearn.feature_extraction.text import CountVectorizer
bow = CountVectorizer(stop_words="english")
```

```
text_data = ["I am interested in NLP",
             "This is a good book with good topic",
             "Feature extraction is very important topic"]
```

```
print("doc1: ", len(text_data[0]))
print("doc2: ", len(text_data[1]))
print("doc3: ", len(text_data[2]))
```

```
doc1: 22
doc2: 35
doc3: 42
```

```
bow.fit(text_data)

<
  CountVectorizer
CountVectorizer(stop_words='english')
>
```

```
feature_name =bow.get_feature_names_out()
feature_name

array(['book', 'extraction', 'feature', 'good', 'important', 'interested',
      'nlp', 'topic'], dtype=object)
```

```
bow_features = bow.transform(text_data)
bow_features

<Compressed Sparse Row sparse matrix of dtype 'int64'
with 9 stored elements and shape (3, 8)>
```

```
bow_features_array = bow_features.toarray()
print(bow_features_array)

[[0 0 0 0 0 1 1 0]
 [1 0 0 2 0 0 0 1]
 [0 1 1 0 1 0 0 1]]
```

```
print(bow.get_feature_names_out())

['book' 'extraction' 'feature' 'good' 'important' 'interested' 'nlp'
 'topic']
```

```
for sentences, feature in zip(text_data, bow_features_array):
    print(sentences)
    print(feature)

I am interested in NLP
[0 0 0 0 0 1 1 0]
This is a good book with good topic
[1 0 0 2 0 0 0 1]
Feature extraction is very important topic
[0 1 1 0 1 0 0 1]
```

```
from sklearn.feature_extraction.text import TfidfVectorizer

vectorizer = TfidfVectorizer()
```

```
X = vectorizer.fit_transform(text_data)
analyze = vectorizer.build_analyzer()
print(vectorizer.get_feature_names_out())

for i in range(1,4):
    print(f"Document {i} : {text_data[i-1]}")
    print("Document transform", X.toarray())

['an' 'book' 'extraction' 'feature' 'good' 'important' 'in' 'interested'
 'is' 'nlp' 'this' 'topic' 'very' 'with']
Document 1 : I am interested in NLP
Document 2 : This is a good book with good topic
Document 3 : Feature extraction is very important topic
Document transform [[0.5      0.      0.5      0.      0.      0.      0.
 0.5      0.
 0.      0.
 [0. 0.35013871 0.      0.      0.70027741 0.
 0.      0.      0.26628951 0.      0.35013871 0.26628951
 0.      0.35013871]
 [0.      0.      0.44036207 0.44036207 0.      0.44036207
 0.      0.      0.3349067 0.      0.      0.3349067
 0.44036207 0.      ]]
```

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